MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 11 · ALGEBRA II · WEEK 7

Pre-Session

Lesson 6-5

Properties of Logarithms

Lesson 6-6

Exponential and Logarithmic Equations

Mr Malek Thiab

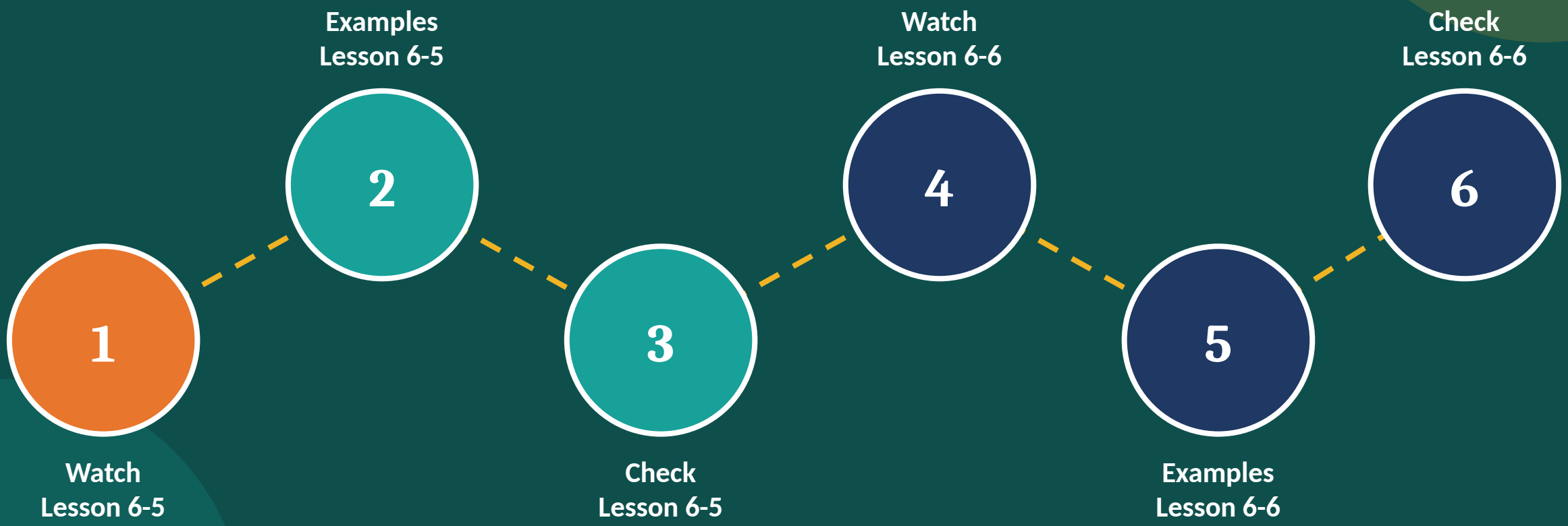
Home task before class

Watch · Study · Check · Submit on the LMS before class

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Your home journey

Six stations before the lesson: watch, study the examples, check yourself, twice.



STATION 1 · WATCH · LESSON 6-5

Properties of Logarithms

WATCH BEFORE CLASS · 5:40



Properties of Logarithms

The Organic Chemistry Tutor · 5:40

Open on YouTube

LISTEN FOR

1. What happens to a logarithm when the number inside is a product, and when it is a quotient.
2. What happens to an exponent that sits on the number inside a logarithm.
3. How the change of base formula writes one logarithm as a ratio of two logarithms.

KEY IDEA

$$\log_b(MN) = \log_b M + \log_b N, \quad \log_b \frac{M}{N} = \log_b M - \log_b N$$

$$\log_b M^p = p \log_b M, \quad \log_b x = \frac{\log x}{\log b}$$

STATION 2 · EXAMPLES · LESSON 6-5

Study the worked examples

EXAMPLE A

1

Expand $\log_2(8x)$

$$\log_2 8 + \log_2 x$$

$$= 3 + \log_2 x$$

EXAMPLE B

2

Evaluate $\log_4 8$ exactly

$$\frac{\log 8}{\log 4} = \frac{3\log 2}{2\log 2}$$

$$= \frac{3}{2}$$

STATION 3 · CHECK · LESSON 6-5

Check yourself: three questions

1

Expand $\log_5 \dfrac{25}{y}$

2

Condense $2\log_3 x + \log_3 5$

3

Evaluate $\log_8 32$ exactly

Done when

you can name the rule you used for each question and write each answer without a calculator.

STATION 4 · WATCH · LESSON 6-6

Exponential and Logarithmic Equations

WATCH BEFORE CLASS · 7:08

Solving Exponential and Logarithmic Equations

Professor Dave Explains · 7:08

Open on YouTube

LISTEN FOR

1. When both sides of an exponential equation can be written with the same base, and what you do next.
2. What you do when the bases cannot match, and where the variable goes.
3. Why a solution to a logarithmic equation must be checked in the original equation.

KEY IDEA

$$b^m = b^n \Rightarrow m = n, \quad a^x = c \Rightarrow x = \frac{\log c}{\log a}$$

$$\log_b x = k \Rightarrow x = b^k, \quad \text{check every answer}$$

STATION 5 · EXAMPLES · LESSON 6-6

Study the worked examples

EXAMPLE A

1

Solve $4^x = 32$

$$(2^2)^x = 2^5$$

$$2x = 5$$

$$x = \frac{5}{2}$$

EXAMPLE B

2

Solve $\log_2(x+1) + \log_2(x-1) = 3$

$$\log_2(x^2 - 1) = 3,\ x^2 - 1 = 8$$

$$x = 3 \text{ or } x = -3$$

$x = -3$ fails the check, so $x = 3$

STATION 6 · CHECK · LESSON 6-6

Check yourself: three questions

1

Solve $27^x = 9$

2

Solve $5^x = 7$ (exact form)

3

Solve $\log x + \log(x-3) = 1$

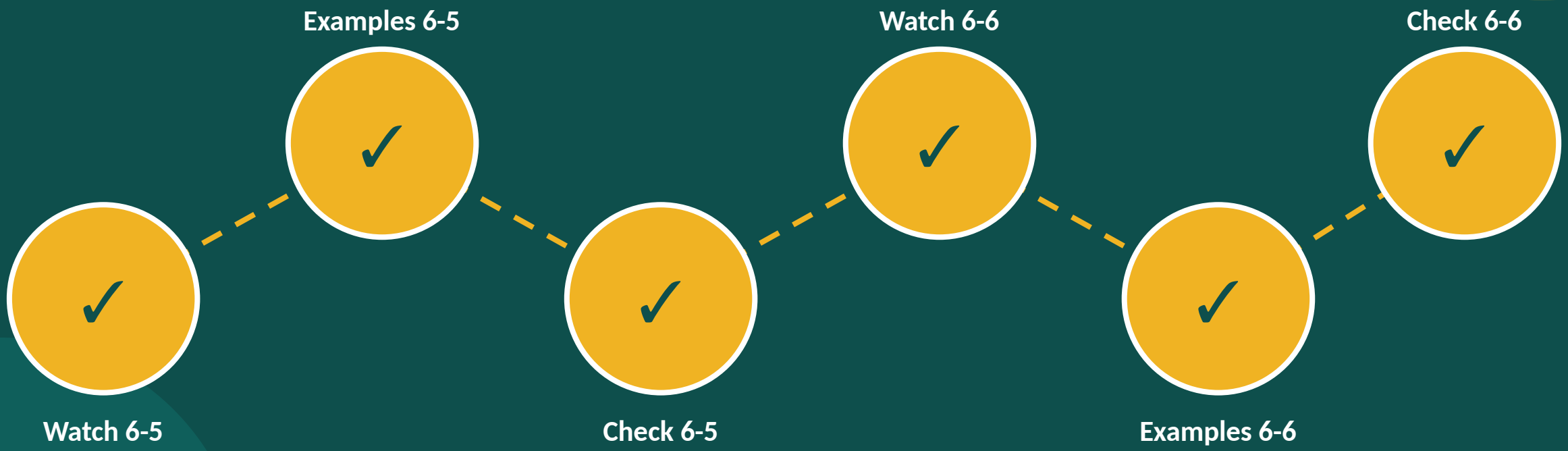
Done when

you have checked every logarithmic answer in the original equation and written each exact answer in simplest form.

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Ready for class

Bring your answers and your questions.



Last week you met logarithms as inverses of exponentials, and you can already switch between exponential form and logarithmic form.

FAITH, RIGHTEOUSNESS AND WISDOM